

MIDTLAEGER, Norway

WMO: 014330

Lat: 59.833N

Lon: 6.983E

Elev: 3547

StdP: 12.91

Time Zone: 1.00 (EUC)

Period: 01-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.1	7.6	-10.7	3.5	20.2	-5.0	4.8	17.0	40.4	15.1	36.8	13.6	14.4	80	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	65.3	53.2	62.1	52.2	59.2	51.0	55.2	62.0	53.8	60.1	52.3	58.2	10.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.0	65.6	57.9	50.5	62.0	56.8	48.9	58.5	55.5	24.6	62.2	23.7	60.2	22.8	58.3	58.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.2	31.3	27.0	1.6	70.8	2.8	2.5	-0.4	72.6	-2.0	74.1	-3.5	75.4	-5.5	77.2
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	21.9	21.4	24.9	32.0	37.7	44.6	51.0	49.2	43.3	35.2	29.1	24.7
	DBStd	11.93	6.91	7.08	7.13	6.20	6.04	5.46	5.80	5.96	5.30	6.31	6.29	6.73
	HDD50	5784	872	802	778	541	382	179	58	91	209	460	627	785
	HDD65	11077	1337	1222	1243	991	845	613	436	489	650	924	1077	1250
	CDD50	183	0	0	0	0	2	17	88	67	9	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.1	14.2	12.0	12.6	10.3	9.1	8.4	8.9	9.5	10.7	11.5	13.3	12.4
Precipitation	PrecAvg	70.7	8.9	6.8	6.5	3.5	3.1	3.3	3.8	4.8	6.1	8.1	7.8	8.2
	PrecMax	97.8	19.5	15.4	19.6	7.3	8.3	7.0	6.3	9.4	11.1	17.5	18.3	16.7
	PrecMin	45.4	0.7	0.4	0.8	1.5	0.2	0.6	1.2	1.1	1.2	1.7	1.3	0.2
	PrecStd	12.8	5.0	4.5	3.9	1.5	1.9	1.8	1.2	1.9	3.0	4.1	4.3	5.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	39.9	43.6	48.1	56.5	65.9	70.7	67.5	60.4	53.2	46.6	39.5
		MCWB	33.9	34.3	36.0	40.1	44.2	51.2	55.5	54.0	50.3	45.7	40.6	34.0
	2%	DB	34.2	35.0	38.7	45.2	52.2	60.4	66.6	63.5	56.0	48.3	43.4	36.5
		MCWB	32.5	31.2	33.0	38.2	43.3	47.3	54.1	53.1	49.1	43.7	39.2	33.1
	5%	DB	32.3	32.9	36.0	42.6	49.4	56.7	63.7	60.6	53.3	46.1	40.4	34.8
		MCWB	31.1	30.5	32.5	37.3	42.0	46.3	53.1	52.7	48.1	42.4	37.8	32.1
	10%	DB	30.9	31.0	33.7	40.3	46.6	53.5	60.4	58.0	51.0	43.8	37.5	33.1
		MCWB	29.6	29.1	31.0	35.5	40.2	44.7	51.8	51.7	46.8	40.9	35.0	31.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	36.1	37.3	42.3	47.0	52.9	57.2	56.0	52.5	48.0	42.2	36.9
		MCDB	35.6	38.8	41.1	45.2	53.0	61.1	66.2	62.4	56.6	51.4	45.2	37.7
	2%	WB	32.8	32.9	34.7	39.7	45.1	50.2	55.8	54.8	50.7	45.0	40.3	34.8
		MCDB	33.5	34.0	37.1	43.4	50.6	58.2	63.7	60.7	54.8	47.6	42.4	35.6
	5%	WB	31.3	30.9	33.1	37.8	43.0	47.6	54.6	53.5	49.1	43.2	37.8	33.0
		MCDB	31.9	32.2	35.4	41.7	47.9	54.2	61.6	59.3	52.8	46.0	40.1	34.1
	10%	WB	29.8	29.2	31.3	36.1	40.9	45.8	53.1	52.2	47.4	41.1	35.3	31.3
		MCDB	30.5	30.6	33.5	39.6	45.8	51.8	59.5	57.7	50.9	43.6	37.2	32.5
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	9.0	9.5	9.1	10.6	9.8	8.1	7.7	7.2	6.9	7.5
		MCDBR	7.8	9.3	11.5	12.1	11.6	15.4	14.4	11.2	10.5	10.3	9.1	8.2
		MCWBR	7.6	7.8	8.1	7.2	7.1	8.9	6.9	5.5	6.3	8.0	7.6	8.0
	5% WB	MCDBR	7.6	8.4	9.7	9.8	10.0	13.3	12.8	10.3	8.9	8.9	8.7	8.0
		MCWBR	7.6	7.8	7.9	6.7	7.0	8.2	6.7	6.2	6.7	7.9	8.1	8.2
Clear-Sky Solar Irradiance	taub	0.207	0.243	0.271	0.301	0.306	0.304	0.318	0.312	0.288	0.268	0.218	0.181	
	taud	2.168	2.278	2.254	2.234	2.328	2.486	2.477	2.486	2.549	2.476	2.221	2.044	
	Ebn at Noon	198	252	277	286	293	295	288	281	272	242	189	158	
	Edh at Noon	13	20	29	36	35	31	30	28	22	17	12	9	
All-Sky Solar Radiation	RadAvg	97	286	694	1275	1494	1568	1438	1127	720	346	126	64	
	RadStd	23	71	145	237	263	193	191	99	77	56	24	9	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page